

NIFTY 100 Financial Intelligence Platform

Analyst Guide & User Documentation

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Purpose. This guide explains how an analyst uses the Streamlit dashboard, stock screener, peer analysis, trend and sector views, capital-allocation intelligence, annual-report repository, PDF tearsheets, and the FastAPI service.

Universe: 92-company NIFTY 100 analytical universe. **Primary interfaces:** Streamlit dashboard and FastAPI REST API. **Storage:** SQLite.

1. Platform Overview

The platform transforms structured financial statement data into a reusable analytics workflow. The implementation covers ETL and data quality validation, financial ratios, CAGR analysis, cash-flow intelligence, screening, composite scoring, peer benchmarking, valuation, NLP-derived qualitative insights, clustering, reporting, and an API layer.

Core outputs include company-level analysis, six screening presets, peer percentile comparisons, valuation flags, cash-flow patterns, 92 company tearsheets, sector reports, a portfolio summary, KMeans cluster labels, and a 16-endpoint FastAPI service.

System architecture follows a layered design: raw ingestion → ETL/normalisation → SQLite storage → analytics → intelligence → reporting → dashboard/API. The original project specification describes this as a seven-layer data platform with 12 analytical modules. ■filecite■turn42file2■L77-L88■

Technology stack: Python, pandas/openpyxl, SQLite, NumPy/scipy/scikit-learn, Plotly/matplotlib, Streamlit, FastAPI/Uvicorn, ReportLab, NLTK/regex and pytest. ■filecite■turn42file2■L89-L109■

2. Starting the Dashboard

From the project root, activate the virtual environment and launch Streamlit:

```
python -m streamlit run src/dashboard/app.py
```

The dashboard opens locally, normally on port 8501. The left navigation provides eight screens: Home, Company Profile, Screener, Peer Comparison, Trend Analysis, Sector Analysis, Capital Allocation, and Annual Reports.

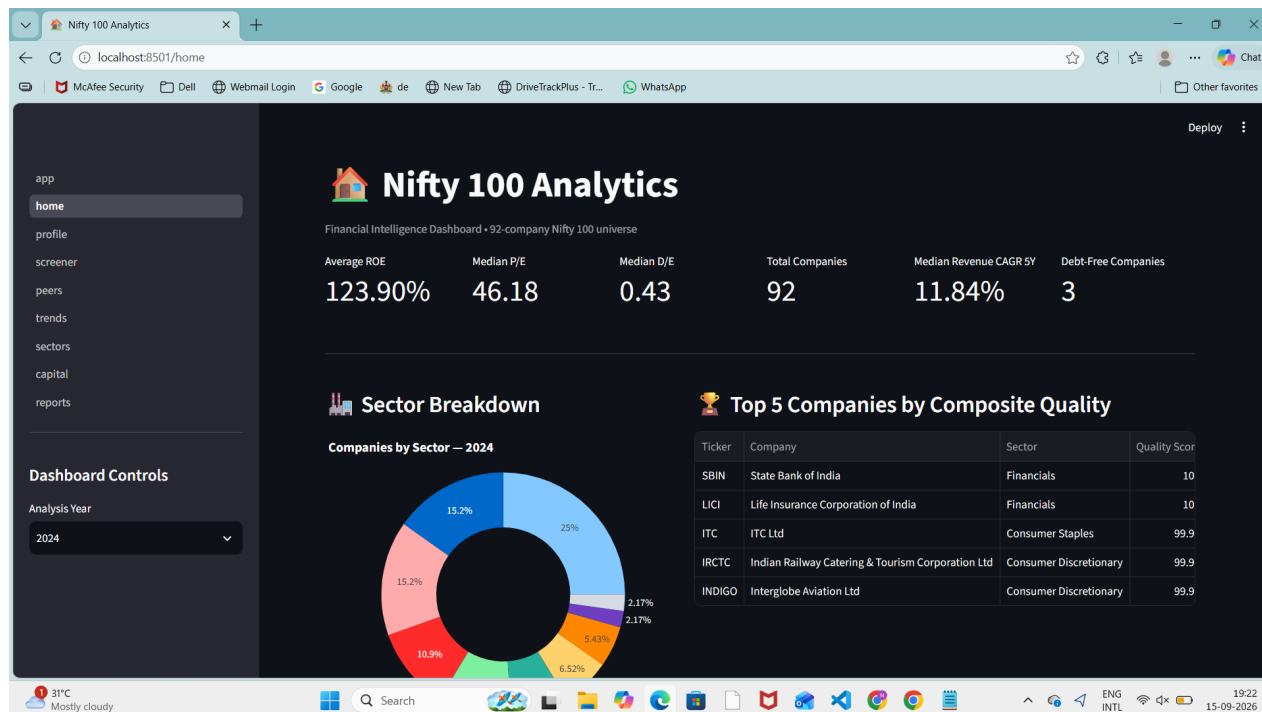


Figure 1. Home dashboard with portfolio-level KPIs, sector breakdown and top composite-quality companies.

The Home screen is the best starting point for a quick portfolio-level view. Review the company count, median valuation/leverage, growth, debt-free count, sector mix and top composite scores before drilling into an individual company.

3. Company Profile

Use the Company Profile screen to investigate one ticker. Search by company name or ticker, then select the company from the dropdown. The screen presents the sector, sub-sector, NSE ticker, website, company description and financial intelligence sections.

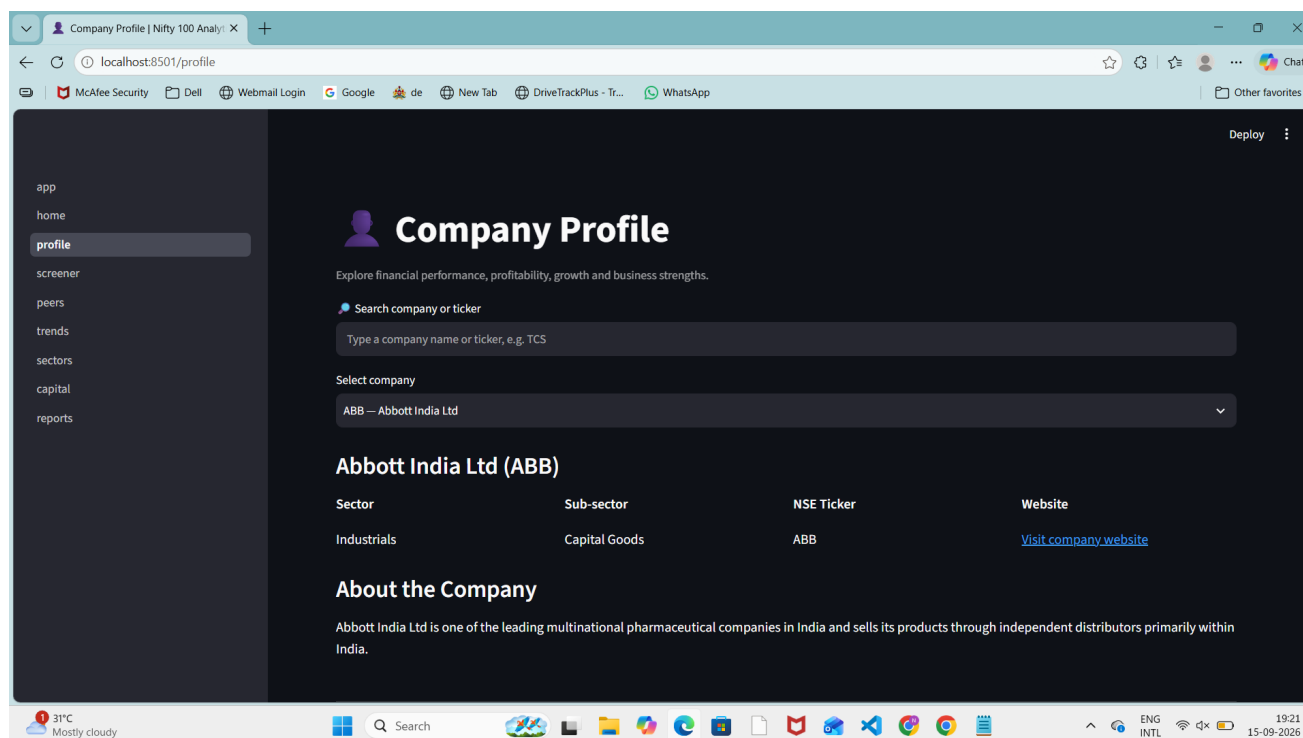


Figure 2. Company Profile screen.

Analyst workflow: (1) search/select a company, (2) review business description and sector context, (3) inspect financial KPIs and historical trends lower on the page, and (4) use the generated tearsheet when a compact offline summary is required.

4. Stock Screener

The Screener applies financial quality, valuation, growth and leverage conditions to the 92-company universe. The sidebar contains preset and custom controls; the result table can be downloaded as CSV.

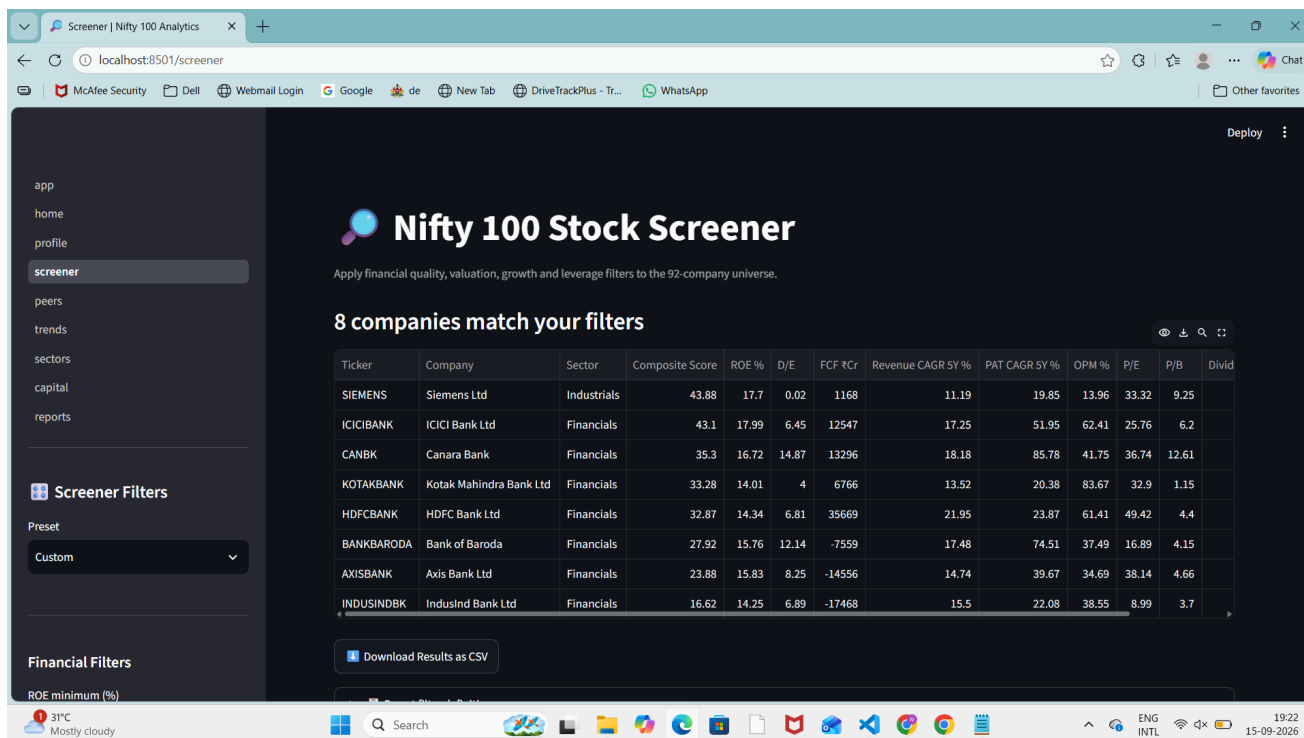


Figure 3. Screener with ranked company results and financial filters.

Preset workflow: choose a named preset such as Quality Compounder, review the returned count, inspect the table columns, and download the result if it will be used outside Streamlit. Custom mode allows analyst-defined thresholds.

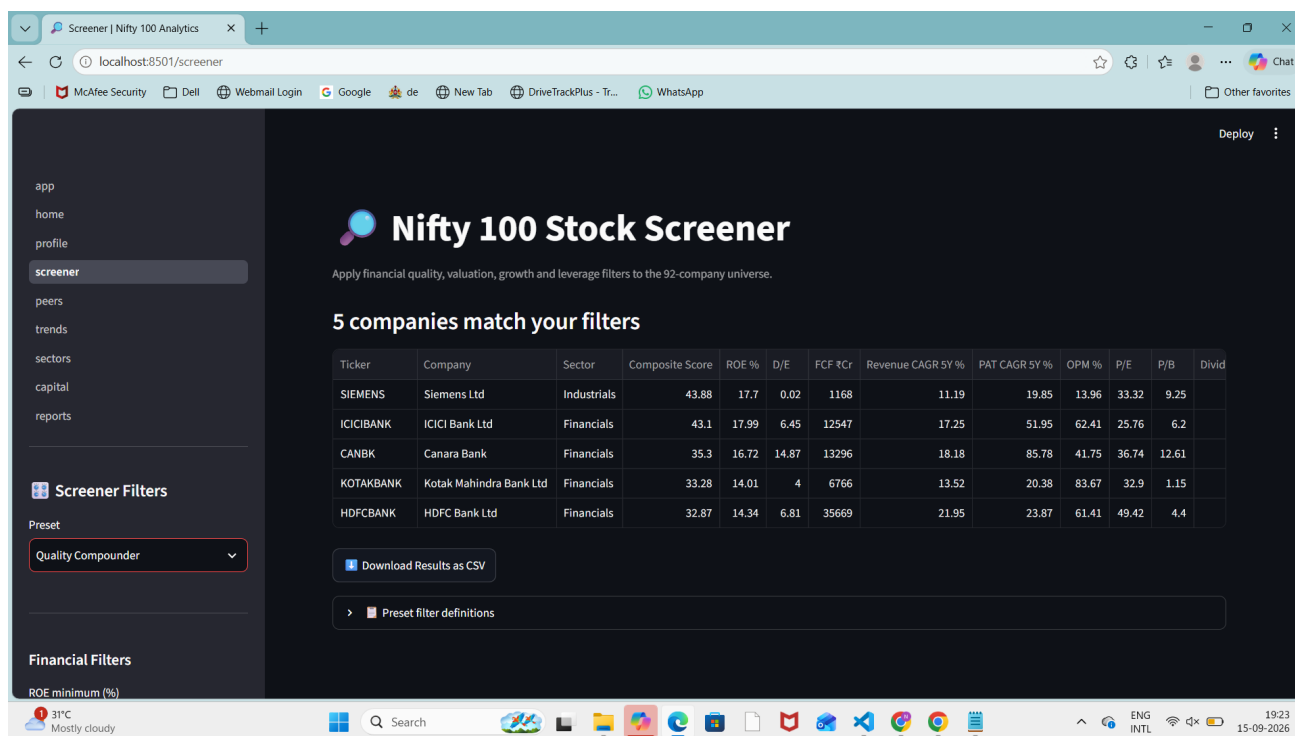


Figure 4. Quality Compounder preset example.

5. Peer Comparison

Peer Comparison benchmarks a selected company against its assigned peer group using percentile-ranked metrics. Select a peer group first, then a company. The radar visual compares the company with the peer benchmark.

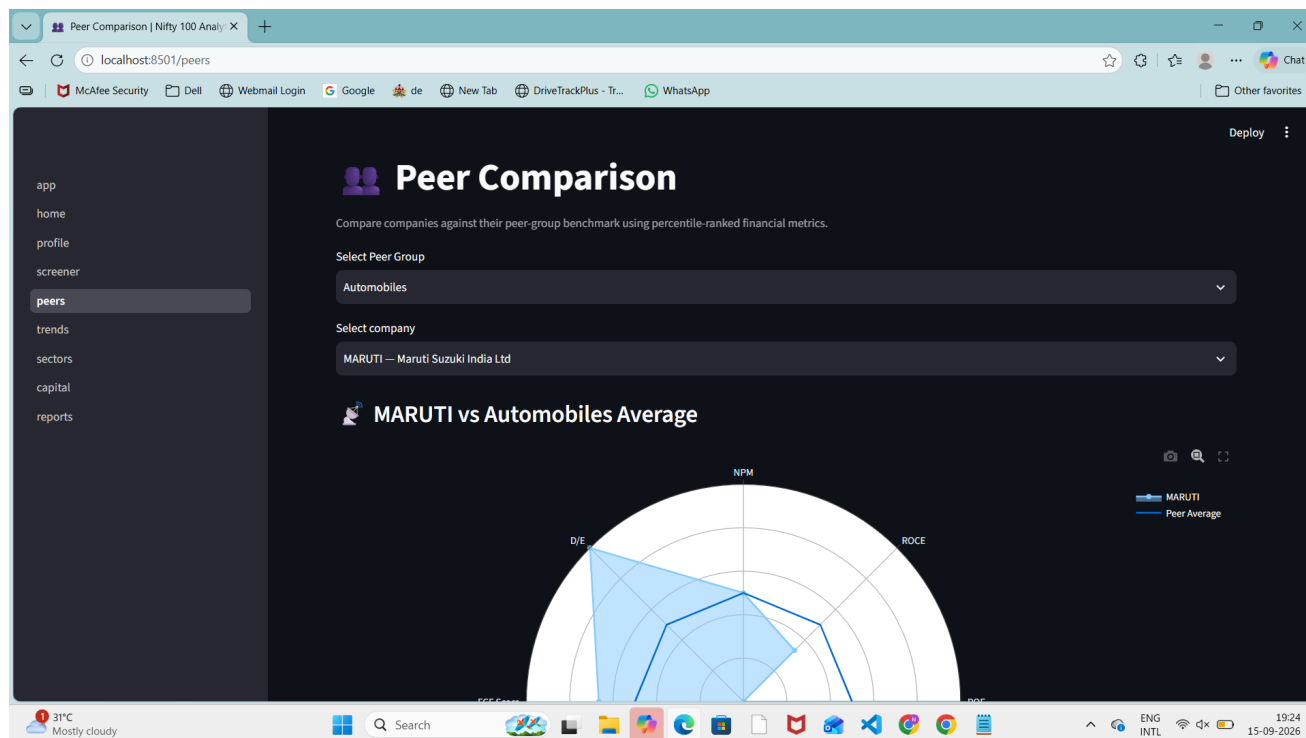


Figure 5. Peer Comparison radar view.

Use this screen when the question is relative rather than absolute: for example, whether a company's profitability, cash flow, growth or leverage is stronger or weaker than comparable businesses. The project specification defines 11 peer groups and percentile-based comparison as a core module. ■filecite■turn43file2■L120-L148■

6. Trend Analysis

Trend Analysis provides up to 10 years of historical KPI movement. Search/select a company and choose up to three metrics. Use the chart to identify sustained improvement, deterioration, cyclicity or inflection points.

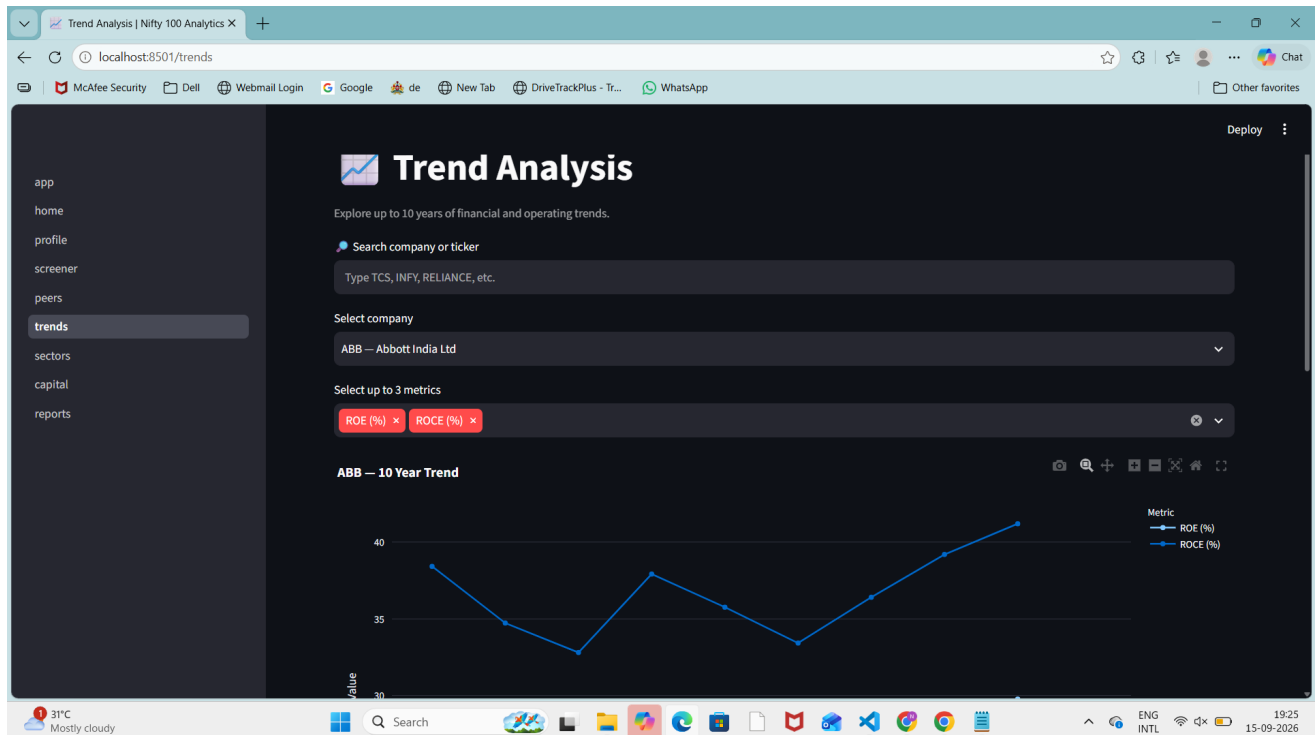


Figure 6. Ten-year trend analysis example.

Recommended metrics include ROE, ROCE, operating margin, net margin, asset turnover, debt-to-equity and other available KPIs. Treat missing years as missing observations rather than assuming that a flat line represents unchanged performance.

7. Sector Analysis

Sector Analysis lets analysts compare companies within a broad sector and inspect sector-level median KPIs. Select a sector from the dropdown. Some sectors may show a data-availability message when the fields required for a visual are insufficient.

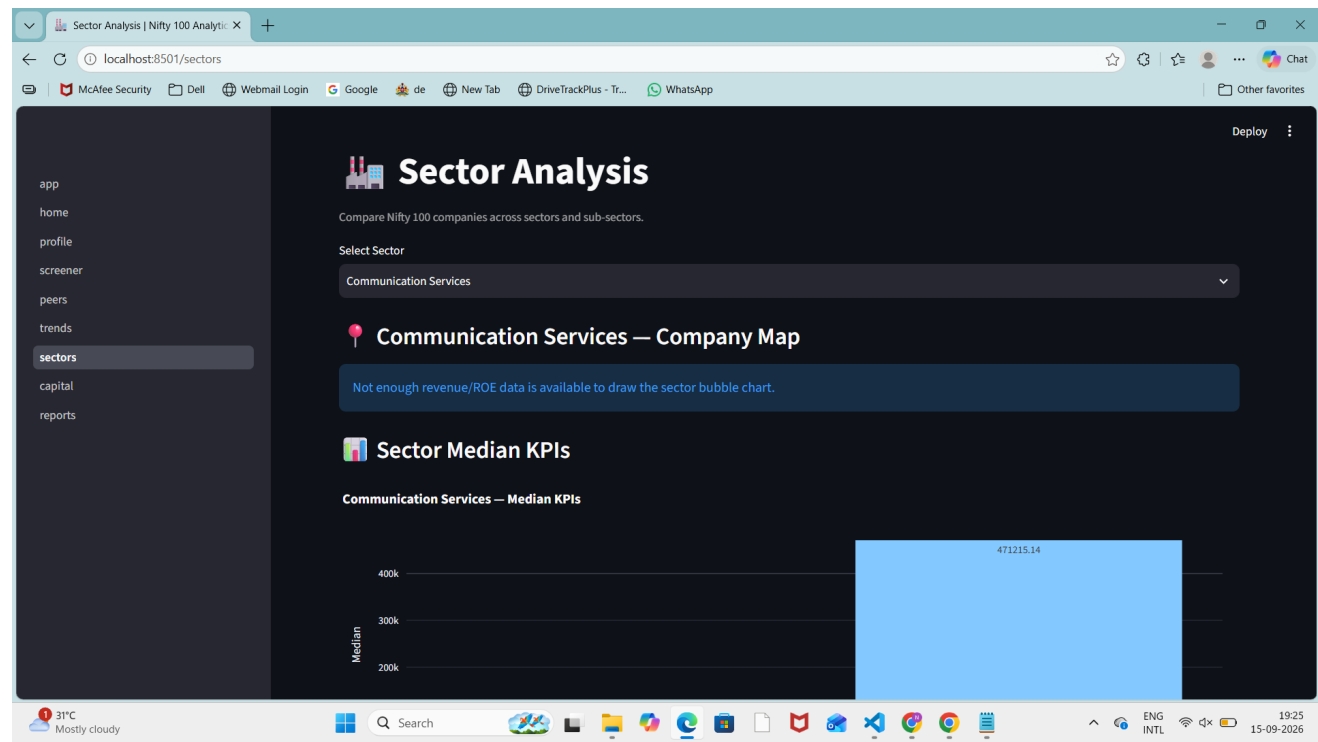


Figure 7. Sector Analysis example and data-availability handling.

The project specification separates broad sector and sub-sector mappings and expects sector-relative analytics.

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8. Capital Allocation & Annual Reports

Capital Allocation classifies the latest company-level pattern into categories such as reinvestment, cash generation, dividend payout and capital stress. The visual map makes portfolio-wide capital-allocation patterns easy to compare.

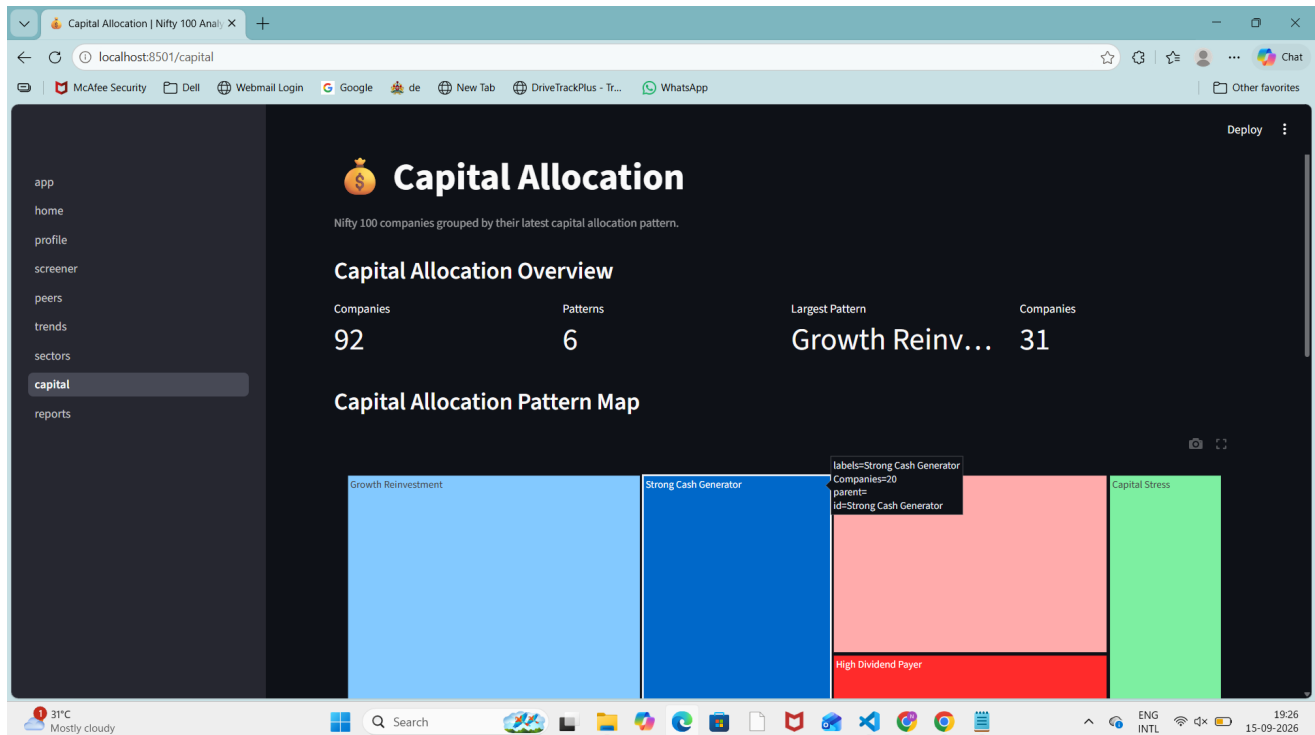


Figure 8. Capital Allocation pattern map.

Annual Reports provides available report links for a selected company. Open a listed BSE/PDF report in a new browser tab when primary-source document review is needed.

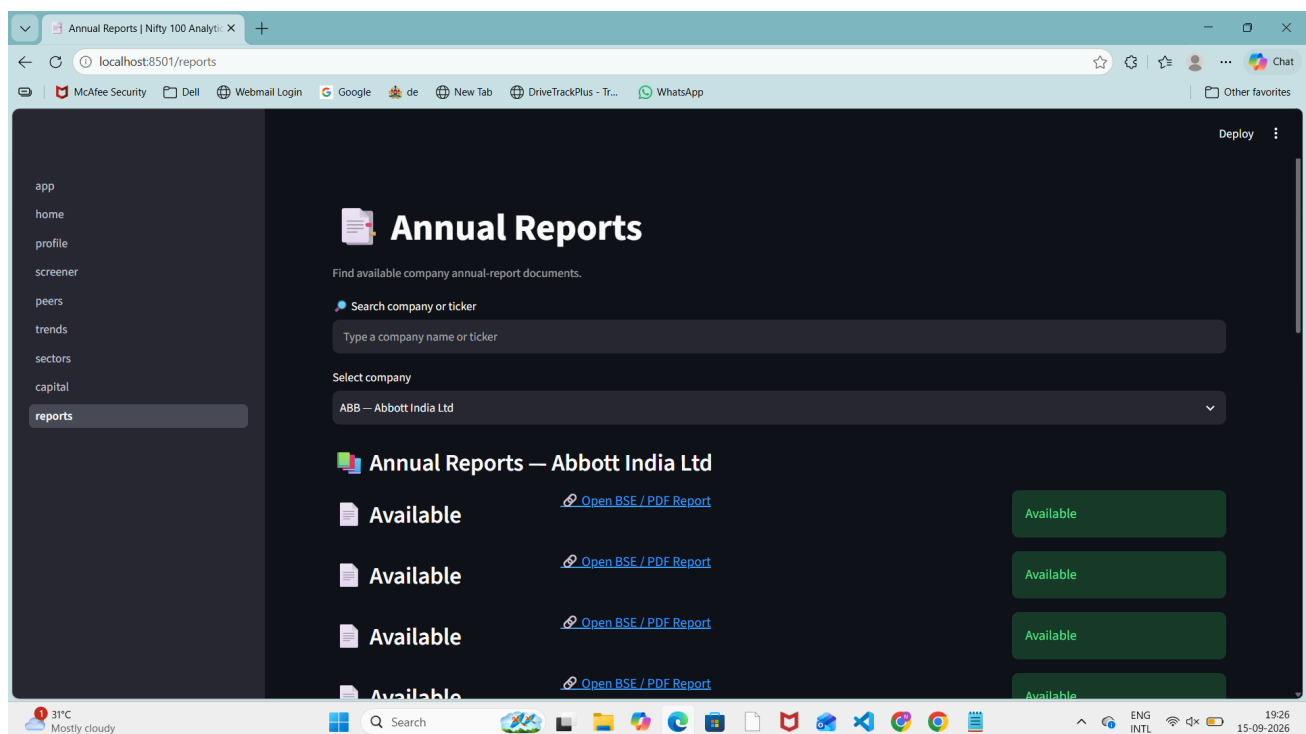


Figure 9. Annual Reports screen.

9. Generating PDF Tearsheets

Company tearsheets are generated in the reporting layer and stored under reports/tearsheets/. The project target is one two-page tearsheet for each of the 92 companies. The original specification defines D-16 as 92 two-page company PDFs. ■filecite■turn43file2■L139-L148■

Typical generation command from the project root:

```
python -m src.reports.tearsheet
```

If the repository uses a dedicated script instead, run the corresponding report-generation script documented in the current README. The generated files should be inspected visually for long text, missing values, clipped tables and page overlap.

Analyst use: open a company's PDF when you need a compact snapshot for review, sharing internally, or attaching to an analysis pack. The PDF is a derived analytical output; primary-source filings remain the appropriate reference for original financial disclosures.

10. FastAPI / Swagger

Launch the API from the project root:

```
$env:PYTHONPATH=(Get-Location).Path
python -m uvicorn src.api.main:app --host 127.0.0.1 --port 8000
```

Open the Swagger UI at the local /docs route. The API exposes company data, P&L, balance sheet, cash flow, ratios, tearsheets, screener results, sector data, peers, market-cap history, portfolio statistics, documents and health information. The project specification defines 16 endpoints. ■filecite■turn42file8■L333-L359■

Example curl requests:

```
curl http://127.0.0.1:8000/api/v1/health
curl http://127.0.0.1:8000/api/v1/companies
curl http://127.0.0.1:8000/api/v1/companies/TCS/ratios
curl "http://127.0.0.1:8000/api/v1/screener"
curl http://127.0.0.1:8000/api/v1/portfolio/stats
```

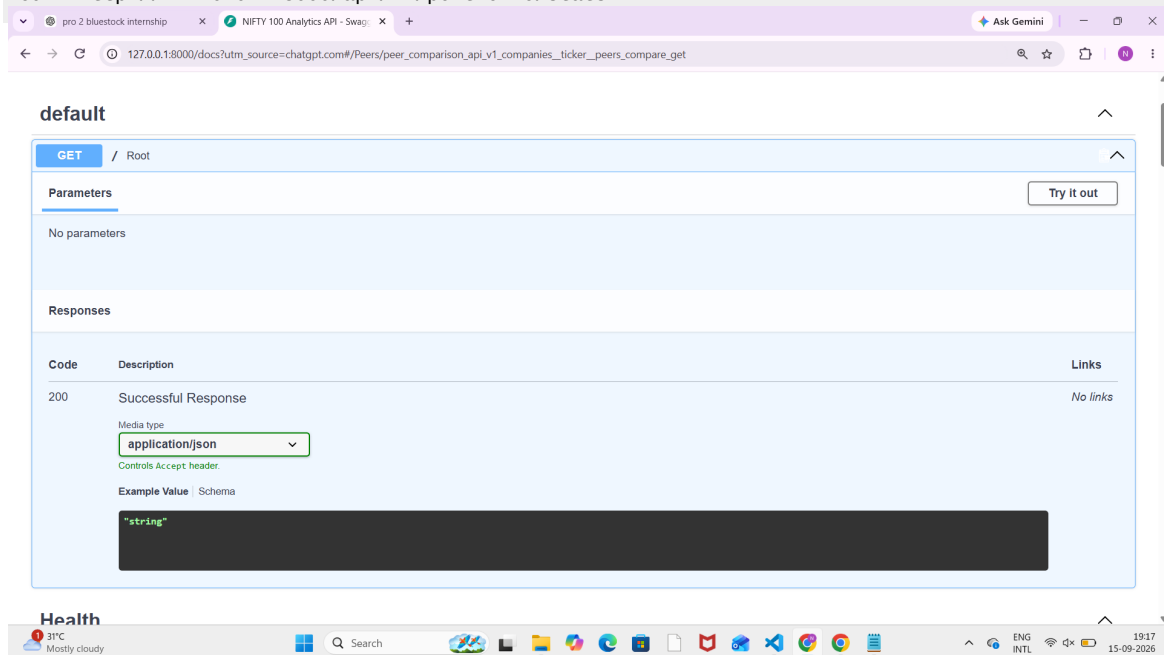


Figure 10. FastAPI Swagger UI.

11. Troubleshooting

Dashboard does not start: confirm the virtual environment is active, dependencies are installed, and the command is executed from the project root.

API import/module error: set PYTHONPATH to the project root before starting Uvicorn.

Port 8000 or 8501 is already in use: stop the existing process or choose another port; keep the chosen URL consistent with the dashboard configuration.

Missing/empty chart: check whether the selected company/sector has enough source data for the requested metric. The dashboard is designed to show a data-availability message rather than fabricate observations.

Missing annual report: the documents dataset contains partial coverage, so an unavailable link should be treated as a data limitation rather than an application error. The original data-relationship specification records documents as partial coverage. ■filecite■turn42file6■L274-L287■

Unexpected screener count: verify the selected preset and custom filters. Presets are configuration-driven, while custom filters can change the returned population.

Tests fail: run pytest from the project root and read the first failing test; do not modify source data merely to make a test pass.

12. Analyst Best Practices

1. Start with the Home screen for context, then drill into a company or peer group.
2. Use multiple KPIs rather than a single ratio to form an investment view.
3. Compare companies within the correct sector/peer group before drawing relative conclusions.
4. Treat simulated market-cap/price inputs as demonstration data where labelled; do not present them as live market data.
5. Treat missing data explicitly; do not infer missing years.
6. Use the tearsheet for summary and primary filings for source verification.
7. Export screener results when an auditable snapshot is required.

The platform's original risk register specifically highlights missing historical years, broken report URLs, parser-format changes, financial-company leverage interpretation, negative-equity ROE, simulated datasets, PDF overflow, dashboard performance and partial peer coverage as operational risks. ■filecite■turn42file7■L303-L320■

Documentation status: this guide covers the required Day 44 topics—screener usage, navigation of dashboard screens, PDF tearsheet generation, API examples and troubleshooting. The project specification explicitly requires the guide to be at least 10 pages. ■filecite■turn43file3■L236-L244■